



SJAA EPHEMERIS

SJAA Activities Calendar

Jim Van Nuland

May (late)

- 21 Houge Park star party. Sunset 8:15 p.m., 65% moon sets 2:21 a.m. Star party hours: 9:15 until midnight.
- 29 **General Meeting** at 8 p.m. Our speaker is Dr. Bradford Holden of Lick Observatory, speaking on The Physical Evolution of Massive Galaxies.

June

- 4 Astronomy Class at Houge Park. 7:30 p.m. Topic is TBA.
- 4 Houge Park star party. Sunset 8:24 p.m., 47% moon rises 1:26 a.m. Star party hours: 9:30 until midnight.
- 5 Dark Sky weekend. Sunset 8:25 p.m., 37% moon rises 1:51 a.m.
- 12 Dark Sky weekend. Sunset 8:29 p.m., No moon. Henry Coe Park's "Astronomy" lot has been reserved.
- 18 Houge Park star party. Sunset 8:31 p.m., 51% moon sets 12:54 a.m. Star party hours: 9:45 until midnight.
- 29 **General Meeting** at 8 p.m. Our speaker is Brian Day of NASA/Ames, who will speak on the LCROSS mission.

July

- 2 Astronomy Class at Houge Park. 7:30 p.m. Topic is TBA.
- 2 Houge Park star party. Sunset 8:32 p.m., 63% moon rises 11:53 p.m. Star party hours: 9:45 until midnight.
- 10 Dark Sky weekend. Sunset 8:30 p.m., No moon. Henry Coe Park's "Astronomy" lot has been reserved.
- 16 Houge Park star party. Sunset 8:27 p.m., 37% moon sets 11:27 p.m. Star party hours: 9:30 until midnight.
- 24 **General Meeting** at 8 p.m. Our speaker is T.B.A.,
- 30 Astronomy Class at Houge Park. 7:30 p.m. Topic is TBA.
- 30 Houge Park star party. Sunset 8:17 p.m., 77% moon rises 10:21 p.m. Star party hours: 9:15 until midnight

The Board of Directors meets before each general meeting at 6:30 p.m. All are welcome to attend.

24 hour news and information hotline:

(408) 559-1221

<http://www.sjaa.net>

Dark Matter and Dark Energy: How Embarrassing!

Paul Kohlmler

Has it ever occurred to you that the main thing that happens when science advances is that we end up with more questions than we had before? For example, what is the stuff of the universe? Matter, at least matter as we know it, makes up only 4% of the universe. The rest is dark matter or dark energy and we barely know anything about them. How embarrassing!

It used to be a lot easier. The universe had matter and energy and then Einstein showed $E=Mc^2$. That meant that matter could be turned into energy. This is how the sun creates energy, fusing hydrogen into helium and some leftovers to create energy.

Fritz Zwicky measured the mass of a galaxy cluster in the 1930's. He noticed that the luminous matter in the galaxy cluster was far less than his measurements. Then, in 1959, Louise Volders noted that galaxy M33 does not rotate in a Keplerian manner. The speed of objects should be slower if the objects are farther from the center. In the 1970's Vera Rubin found many galaxies that had this kind of rotation. The most likely reason for this was that there was more mass in the galaxy than was apparent particularly in the galaxy's halo. In fact, if that extra mass isn't there, the rotation of galaxies would pull them apart. This dark matter never heats up enough to radiate in the infrared so it is dark and it must be very cold. So this stuff is called cold, dark matter.

Dark matter turned out to be useful for describing how the universe could have structure like the galactic clusters when the Big Bang might otherwise have created a bland, homogenous universe. Many candidates for dark matter have been offered. The most popular is a particle called a Higgs boson. This particle would be much larger than a proton or neutron but it breaks down into smaller particles in the presence of heat. The Large Hadron Collider (LHC) in Europe may provide evidence that the Higgs boson is the right guy.

Unfortunately, while waiting for confirmation on the

Continued on page 4

Saturn's swirling snowstorm

Akkana Peck

Saturn is beautifully placed for viewing, up high at nightfall, with rings still tilted nearly edge-on to us: they're tilted only about 1.7 degrees.

Last month I wrote about a few of Saturn's plethora of moons. But moons and rings aren't all there is to see on Saturn: you can watch some weather, too.

Weather on Saturn is a lot more subtle than on Jupiter. I remember, as a kid, looking through the Jason 60mm refractor I'd found at a garage sale and seeing Jupiter's equatorial bands and polar regions. You need a bit more equipment than that to see weather on Saturn.

But recently we got a reminder of the value of amateur Saturnian weather-watching. In March, noted amateur astrophotographer Christopher Go, using his C-11 in the Philippines, shot an image of Saturn that showed a new and massive blizzard gaining strength. Go and Australian amateur Anthony Wesley emailed the Cassini project with pictures of the storm.

The Cassini team was able to image the area a few weeks later

with the spacecraft's composite infrared spectrometer. They found unexpectedly high levels of phosphine being pushed up from lower levels of Saturn's atmosphere, indicating powerful updrafts driving the storm. Cassini scientists say there's probably a fierce ammonia-ice blizzard about 100 km below the bottom of Saturn's stratosphere, probably driven by fierce storms another 100 to 200 km lower, involving lightning and clouds of water and ammonia.

Somehow that makes the drizzle as I'm writing this column seem a lot less significant.

This isn't the first time in the spotlight for either Go or Wesley. Go was active in imaging Jupiter's "Spot Jr" a few years ago, and has even worked with the Hubble Heritage team imaging a quadruple Saturn moon transit, while Wesley was in the news last summer for discovering a new dark spot on Saturn.

For the March Saturn storm, they were responding to an alert that went out in February that Cassini's radio and plasma wave instrument was showing increased static, possibly indicating a large storm.

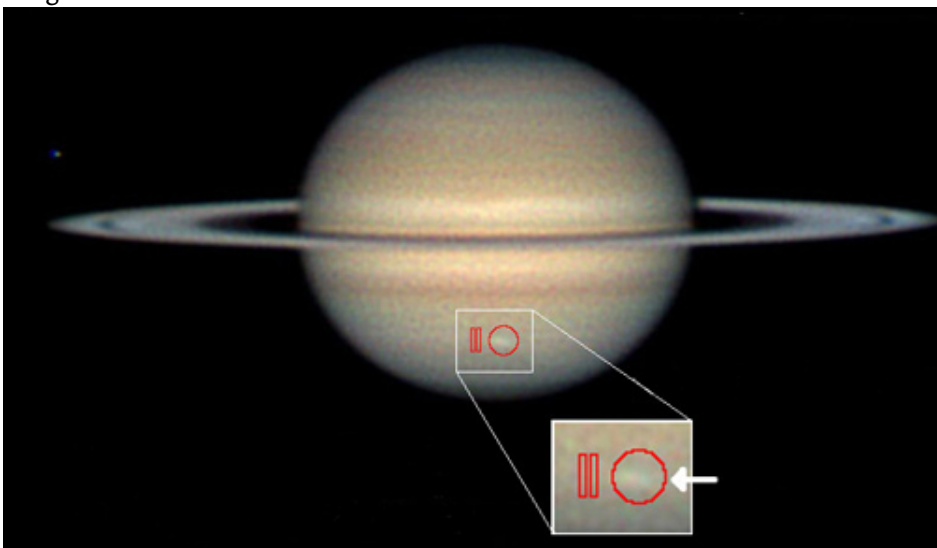
So how should amateurs get involved? Try joining some of the active Saturn observing lists — perhaps some of the ALPO lists on Yahoo Groups — and see what hot new feature everybody's talking about.

If you do imaging, consider making a web page showing how your images change from day to day. Remember, scientists get a lot of amateur images submitted, and they aren't looking for pretty images so much as how features on the planet are changing. Go himself has a great example of that sort of page, including details about his observing setup, at: <http://astro.christone.net/saturn/>

But enough about Saturn. Early on June evenings, before you swing up to Saturn, check out Mars and Venus in the western sky. Venus stays low, while Mars is in the belly of Leo.

Pluto is at opposition on June 25th, so this marks the beginning of Pluto season. Unfortunately, it's located in a very tough field — northern Sagittarius, right in the heart of the Milky Way. That means lots of guide stars, but it also means lots of stars to confuse with faint Pluto. As if that weren't enough, it's not just any old central Milky Way location, but right on the edge of M24, the Sagittarius star cloud. And at magnitude 14, Pluto is right about the same magnitude as a lot of those M24 stars. So go after Pluto — but make sure you have a very good chart of the M24 region prepared first.

Jupiter, Uranus and Neptune are morning objects all this month, with Mercury making a morning appearance at the beginning of June but disappearing behind the sun a few weeks later. Early in the month, Jupiter and Uranus make a close pass, getting within half a degree on the morning of the 6th.



Amateur astronomer Christopher Go took this image of the storm on March 13, 2010. The arrow indicates the location of the storm and the red outlines show where Cassini's composite infrared spectrometer gathered data. Image credit: C. Go and NASA/JPL-Caltech/GSFC

Australopithecus squinted at the blue African sky. He had never seen a star in broad daylight before, but he could see one today. Was it dangerous? He stared for a long time, puzzled, but nothing happened, and after a while he strode across the savanna unconcerned.

Millions of years later, we know better.

That star was a supernova, one of many that exploded in our corner of the Milky Way around the Pliocene era of pre-humans. Australopithecus left no records; we know the explosions happened because their debris is still around. The Solar System and everything else within about 300 light-years is surrounded by supernova exhaust—a haze of million-degree gas that permeates all of local space.

Supernovas are dangerous things, and when one appears in the daytime sky, it is cause for alarm. How did Earth survive? Modern astronomers believe the blasts were too far away (albeit

not by much) to zap our planet with lethal amounts of radiation. Also, the sun's magnetic field has done a good job holding the hot gas at bay. In other words, we lucked out.

The debris from those old explosions has the compelling power of a train wreck; astronomers have trouble tearing their eyes away. Over the years, they've thoroughly surveyed the wreckage and therein found a mystery—clouds of hydrogen and helium apparently too fragile to have survived the blasts. One of them, whimsically called "the Local Fluff," is on the doorstep of the Solar System.

"The observed temperature and density of the Fluff do not provide enough pressure to resist the crushing action of the hot supernova gas around it," says astronomer Merav Opher of George Mason University. "It makes us wonder, how can such a cloud exist?"

NASA's Voyager spacecraft may have

found the answer.

NASA's two Voyager probes have been racing out of the solar system for more than 30 years. They are now beyond the orbit of Pluto and on the verge of entering interstellar space. "The Voyagers are not actually inside the Local Fluff," explains Opher. "But they are getting close and can sense what the cloud is like as they approach it."

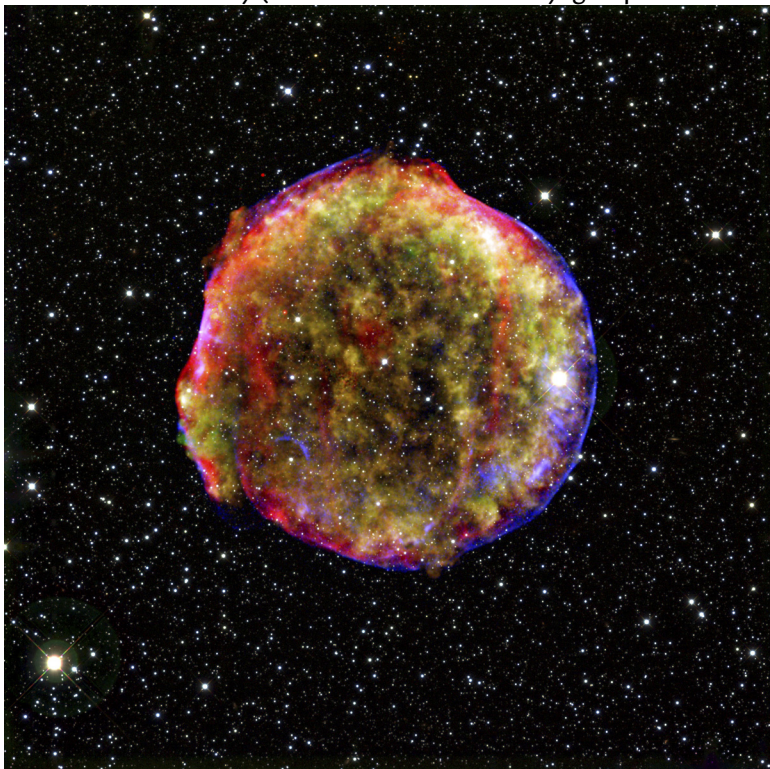
And the answer is....

"Magnetism," says Opher. "Voyager data show that the Fluff is strongly magnetized with a field strength between 4 and 5 microgauss. This magnetic field can provide the pressure required to resist destruction."

If fluffy clouds of hydrogen can survive a supernova blast, maybe it's not so surprising that we did, too. "Indeed, this is helping us understand how supernovas interact with their environment—and how destructive the blasts actually are," says Opher.

Maybe Australopithecus was on to something after all.

Opher's original research describing Voyager's discovery of the magnetic field in the Local Fluff may be found in *Nature*, 462, 1036-1038 (24 December 2009). The Space Place has a new Amazing Fact page about the Voyagers' Golden, with sample images and sounds of Earth. After all, just in case one of the Voyager's ever meets up with ET, we will want to introduce ourselves. Visit <http://spaceplace.nasa.gov/en/kids/voyager>.

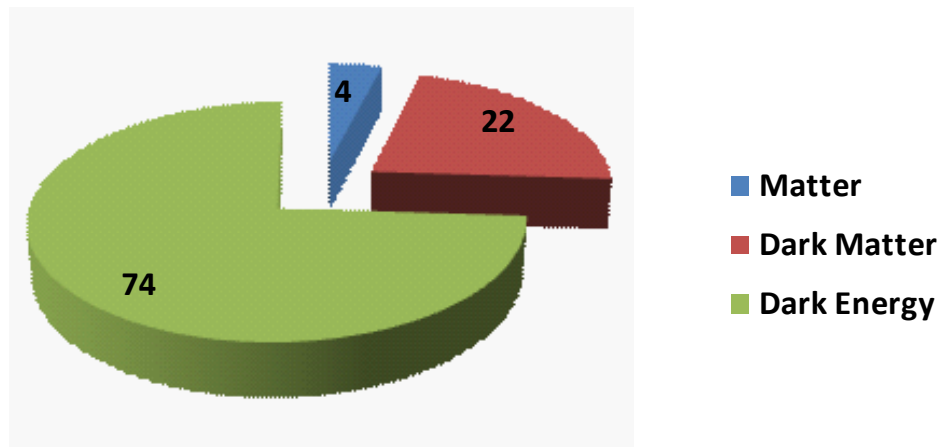


Left-over cloud from the Tycho supernova, witnessed by Tycho Brahe and other astronomers over 400 years ago. This image combines infrared light captured by the Spitzer Space Telescope with x-rays captured by the Chandra X-ray Observatory, plus visible light from the Calar Alto Observatory in Spain.

This article was provided by the Jet Propulsion Laboratory, California Institute of Technology, under a contract with the National Aeronautics and Space Administration.

identity of dark matter, a bigger mystery appeared. Edwin Hubble studied the motion of galaxies and plotted their relative velocities on a graph. What he found was the further away the galaxy was the faster it was moving away from us. Why? If you drop a hundred marbles in the middle of a basketball court, the marbles that are furthest away must have higher velocities - at least in the first few seconds after the drop. Eventually even the furthest marbles would slow down and astronomers wanted to see if the galaxies were slowing down. If they were, then maybe the galaxies would once again collapse to a single point.

But astronomers found that the expansion of the universe was actually accelerating. Dark matter was embarrassing enough but at least it was acting like matter in one way, it had gravity. But whatever was causing this acceleration was doing the opposite.



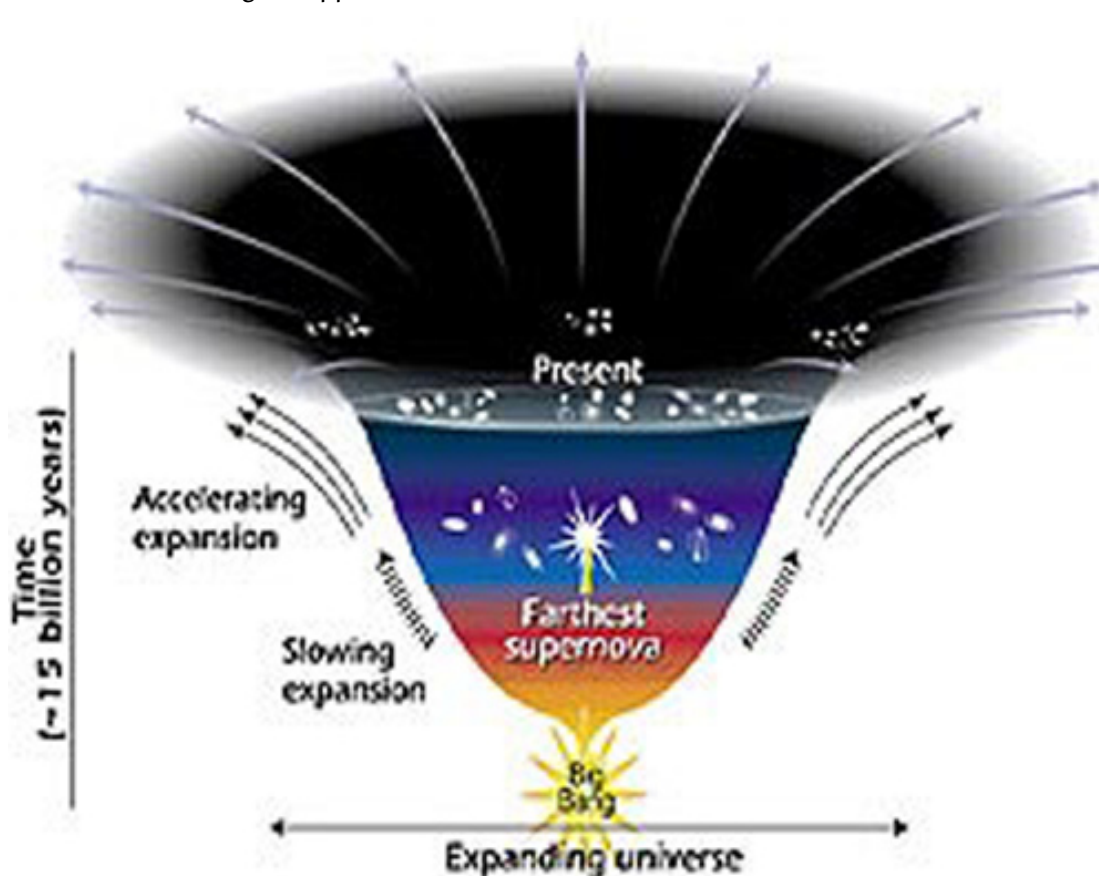
The chart shows the relative abundances of matter, dark matter and dark energy. The numbers shown are from one estimate but the numbers vary from study to study.

This other mystery component was given the name of dark energy. Dark energy itself is constant but it dominates as the universe expands and gravity becomes diluted.

components of the universe. But the total mass of dark matter is probably 4 times the amount of normal matter. And dark energy makes up 5 times the

It would be one thing if dark matter and dark energy were minor

Dark Matter is embarrassing enough but at least it follows the laws of gravity. Dark energy seems to be doing the opposite!



Graphic from NASA

amount of normal and dark matter combined. And to date, we can say very little about either one. How embarrassing!

[Ed.note: In mid-May 2010, a story appeared announcing that missing matter has been found. This missing matter is normal or baryonic matter found in intergalactic space and discovered in X-rays. This matter is also called WHIM (Warm-hot intergalactic matter). This does not change the current estimates of dark matter or dark energy. See <http://sci.esa.int/science-e/www/object/index.cfm?fobjectid=47026> and http://www.sciencecodex.com/xray_discovery_points_to_location_of_missing_matter

Directions to Houge Park

Houge (rhymes with "Yogi") Park is in San Jose, near Campbell and Los Gatos. From Hwy. 17, take the Camden Avenue exit. Go east 0.4 miles, and turn right at the light, onto Bascom Avenue. At the next light, turn left onto Woodard Road. At the first stop sign, turn right onto Twilight Drive. Go three blocks, cross Sunrise Drive, then turn left into the park.

From Hwy. 85, take the Bascom Avenue exit. Go north, and turn right at the first traffic light, onto White Oaks Road. At the first stop sign, turn left onto Twilight Drive. You will now be passing the park. Turn right at the first driveway, into the parking lot.

Observatory Fund

The SJAA Board of Directors has released the following statement:

"The SJAA Board Of Directors has dissolved the Observatory Committee and plans to absorb Observatory Fund donations into the club's General Funds. This ad will be run in the May, June and July Ephemeris seeking input from any donors regarding these plans. If you are a donor and would like to comment, please contact the Board of Directors."

For additional background information, the following information is from last August's SJAA Ephemeris. The author is Greg Claytor.

PUBLIC NOTICE IS HEREBY GIVEN that the San Jose Astronomical Association, a California nonprofit public benefit corporation 501(c)(3), will terminate its effort to obtain or construct an observatory using its Observatory Fund, that it no longer has any plan now or in the future to obtain or construct an observatory and that, without objection, its board of directors will re-direct the clubs Observatory Funds into other acquisitions and/or activities.

If you have contributed to this fund, have knowledge of others who may have contributed to this fund or have any interest at all in this fund, please contact the San Jose Astronomical Association by mail:

San Jose Astronomical Association PO Box 28243 San Jose, CA 95159-8243

Ed. Note: This column will be repeated in the July issue.

Current Astronomical Spacecraft

The following spacecraft are currently being used for astronomical research. In Earth orbit: WISE, SDO, Hubble, MOST, COROT, AKARI, Chandra, XMM-Newton, Suzaku, Rossi XTE, MAXI, Fermi, Integral, Koronas-Foton, TRACE, SORCE, RHESSI, Hinode, ESA Solar package, IZBEX, Resurs-DK.

Beyond Earth Orbit. Orbiting the moon: Lunar Reconnaissance Orbiter. Orbiting the sun: Kepler, Spitzer, STEREO A and B. Heading toward planets: Messenger (to Mercury), New Horizons (to Pluto and the Kuiper Belt). At planets: Spirit, Opportunity, Mars Odyssey, Mars Express, Mars Reconnaissance Orbiter, Venus Express, Cassini. Orbiting a Lagrangian point: SOHO, ACE, WMAP, Planck, Herschel. Flying past asteroids or comets: Hayabusa, Rosetta, EPOXI, Stardust, Dawn. Exiting the solar system: Voyager 1 and 2.

Based on data from Jonathan McDowell. see <http://www.planet4589.org> or Sky & Telescope, February 2010 p.16.

SJAA Yosemite Public Star Party 2010

Jim Van Nuland

The annual SJAA Yosemite star party will be held on September 3, 4, and 5, at Glacier Point in Yosemite National Park. Up to 30 people will be given free admission and camping, in exchange for three public events on Friday, Saturday, and Sunday evenings. The rest of the time we can be tourists.

We are expected to have at least one scope per two people, and to attend at least two of the star parties, not just Friday or Saturday. Sunday is expected to be nearly as busy as Friday and Saturday. For these reasons, this is probably not suitable for a family camping trip.

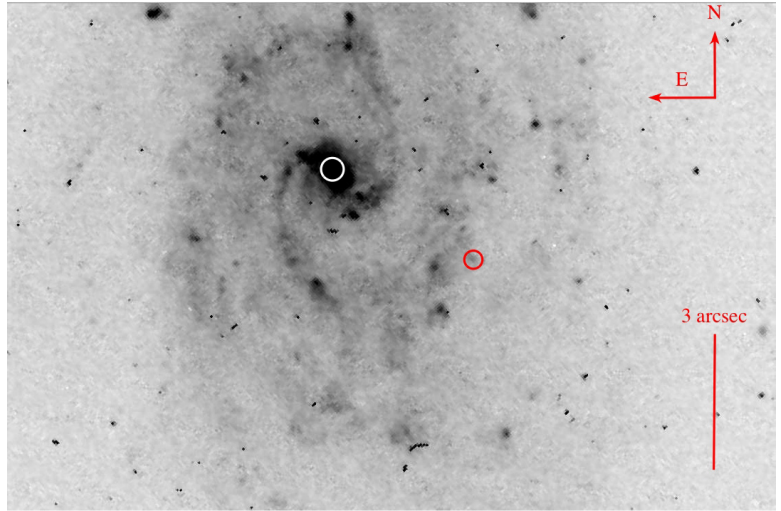
The camping is rough by modern standards: no dining room, no showers, no hot water. Read about it on the SJAA Yosemite page <http://www.sjaa.net/yosemite.shtml>, then contact me with questions. That page also has sun and moon rise and set times.

If you can tolerate the limitations, tell me the number of people you'll have, and the number of scopes that will be set up for the public. E-mail me at jvn@sjpc.org, or phone 408-371-1307 11 a.m. to 11 p. m. Priority is given to SJAA members.

The Last Month In Astronomy

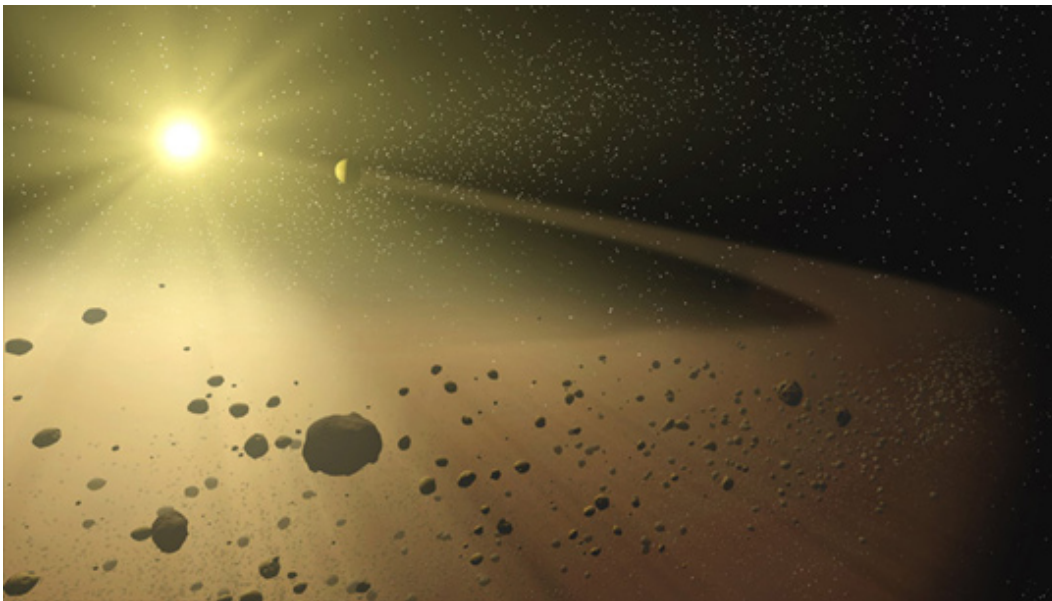
MAY-11-2010 **Black Hole Leaving**

An undergraduate student from the Netherlands has found evidence of a black hole that is moving out of its parent galaxy. Marianne Heida found the supermassive black hole using data from the Chandra X-ray observatory. As gas and dust falls in the black hole it heats up and emits radiation at the high frequencies of X-rays. X-rays also penetrate the other material that is being sucked into black hole which is why black holes appear as nearly pure X-ray emitters. While supermassive black holes (around 1 billion solar masses) are most likely found in galaxy centers, this black hole is being expelled at a high speed. This might be caused by two black holes merging. Computer simulations suggest that the larger resulting black hole will be shot out though it is dependent on starting conditions such as the relative direction and speed of the progenitors. The picture shown here is from Hubble and does not use X-ray data. http://www.sron.nl/index.php?option=com_content&task=view&id=2709&Itemid=588



MAY-11-2010 **A Hole in Space**

The ESA's Herschel spacecraft has found something surprising, a hole in space. Seeing black patches in the sky is nothing new. We know that these areas are dark clouds that are blocking light. Herschel looked at one such patch and found that it was quite dark. But Herschel looks in the infrared which usually finds some radiation from these objects. There are 3 possible explanations: the instruments aren't working (nope, working fine); the cloud is very cold and dense (no, other earth-based instruments can't find anything either); or the cloud is a hole, not a cloud. Says Tom Megeath of the University of Toledo "No one has ever seen a hole like this. It's as surprising as knowing you have worms tunneling under your lawn, but finding one morning that they have created a huge, yawning pit." This hole may have been created by the stellar jets created by stars in NGC 1999. The hole gives us a view into how stars disperse their stellar nursery environment. http://www.esa.int/esaSC/SEMFEAKPO8G_index_0.html



APR-28-2010 **Life Components Found On Asteroid**

"For a long time the thinking was that you couldn't find a cup's worth of water in the entire asteroid belt. Today we know you not only could quench your thirst, but you might just be able to fill up every pool on Earth." So said Don Yeomans of JPL. He was talking about the results of 6 years of measurements of asteroid 24 Themis. These measurements were done by Andrew Rivkin and Joshua Emery using the NASA Infrared Telescope Facility. There, buried in the data, is the signature of

both water ice and organic materials. <http://jpl.nasa.gov/news/news.cfm?release=2010-142>

APR-21-2010 **One planet, Hold the Methane**

NASA's Spitzer Space Telescope has discovered something strange about the planet GJ 436b. This planet is about the size of Neptune making it the smallest exoplanet to have its atmosphere tested. Methane was expected but it hasn't been found. The planet should have everything needed to cook up some of the gas: About 1000 degrees Fahrenheit, atmosphere with hydrogen, carbon and oxygen. Joseph Harrington of the University of Central Florida says "We are scratching our heads". <http://jpl.nasa.gov/news/news.cfm?release=2010-137>

It Must Be Astronomical ...

Loaners

The loaner program offers members a means to try scopes of various sizes and technologies before you buy. For more information please see the loaner program web page: <http://www.sjaa.net/loaners>

Eclipses 2010

Partial Lunar Eclipse - June 26 - visible from San Jose at moonset

Solar Eclipse - July 11 - Not visible from San Jose nor any other part of North America

Total Lunar Eclipse - December 21 - Fully visible from San Jose

Hot Dates

Golden State Star Party, July 10-14, Adin, California. For more information, see <http://www.goldenstatestarparty.org>.

SETIcon - Aug. 13-15, Santa Clara. SETIcon will offer fascinating and fun panels about astrobiology and SETI research, with speakers ranging from SETI Institute scientists to science fiction actors. We'll have four simultaneous program tracks: general sessions, hard-core science, education/family activities, and individual sessions. For more info go to: <http://www.seticon.com/>

SJAA Yosemite Public Star Party 2010 - Sept. 3-5

CalStar - Oct. 7-10. <http://www.observers.org/CalStar/>

Advanced Imaging Conference, October 22-24, Santa Clara. <http://www.aicccd.com/2010/website/index.html>

Q & A

Q: At the center of the galaxy is a SMBH (Super Massive Black Hole). What percentage of the galaxies mass is in this black hole?

A: It is only 0.002% of the galaxies total mass and not likely to eat us up anytime soon. ("Death from the Skies", Phil Plait, pg. 252)

"Who are we? We find that we live on an insignificant planet of a humdrum star lost in a galaxy tucked away in some forgotten corner of a universe in which there are far more galaxies than people." - Carl Sagan

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Members Email Lists:
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Articles for publication should be submitted by the 10th of the previous month. The PDF version is generally available by the 24th of the previous month and the HTML version by the last day of the previous month.

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- ☐ Regular — \$20
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Subscribing to Sky & Telescope magazine through the SJAA saves you \$5 off the regular rate. (S&T will not accept multi-year subscriptions through the club program. Allow 2 months lead time.)

☐ **I prefer to get the Ephemeris newsletter in print form. The newsletter is always available online at <http://ephemeris.sjaa.net>**
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Send e-mail to membership@sjaa.net

Bring this form to any SJAA Meeting or send to the club address (above).
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